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and are present in 6-9 bundles. Each tubule is lined by glandular and ciliated cells and the waste is excreted out through the hindgut. The glandular cells of the malpighian tubules absorb water, salts, and nitrogenous wastes from the haemolymph and transfer them into the lumen of the tubules. The cells of the tubules reabsorb water and certain inorganic salts. By the contraction of the tubules nitrogenous waste is pushed into the ileum, where more water is reabsorbed. It moves into the rectum and almost solid uric acid is excreted along with the faecal matter.

Reproductive system

Cockroach is dioecious or unisexual. They have well developed reproductive organs. The male reproductive system consists of a pair of testes, vasa deferentia, an ejaculatory duct, utricular gland, phallic gland and the external genitalia. A pair of three lobed testes lies on the lateral side of the 4th and 6th abdominal segments. From each testis arises a thin vas deferens, which opens into the ejaculatory duct through the seminal vesicles. The ejaculatory duct is an elongated duct which opens out by the male gonopore

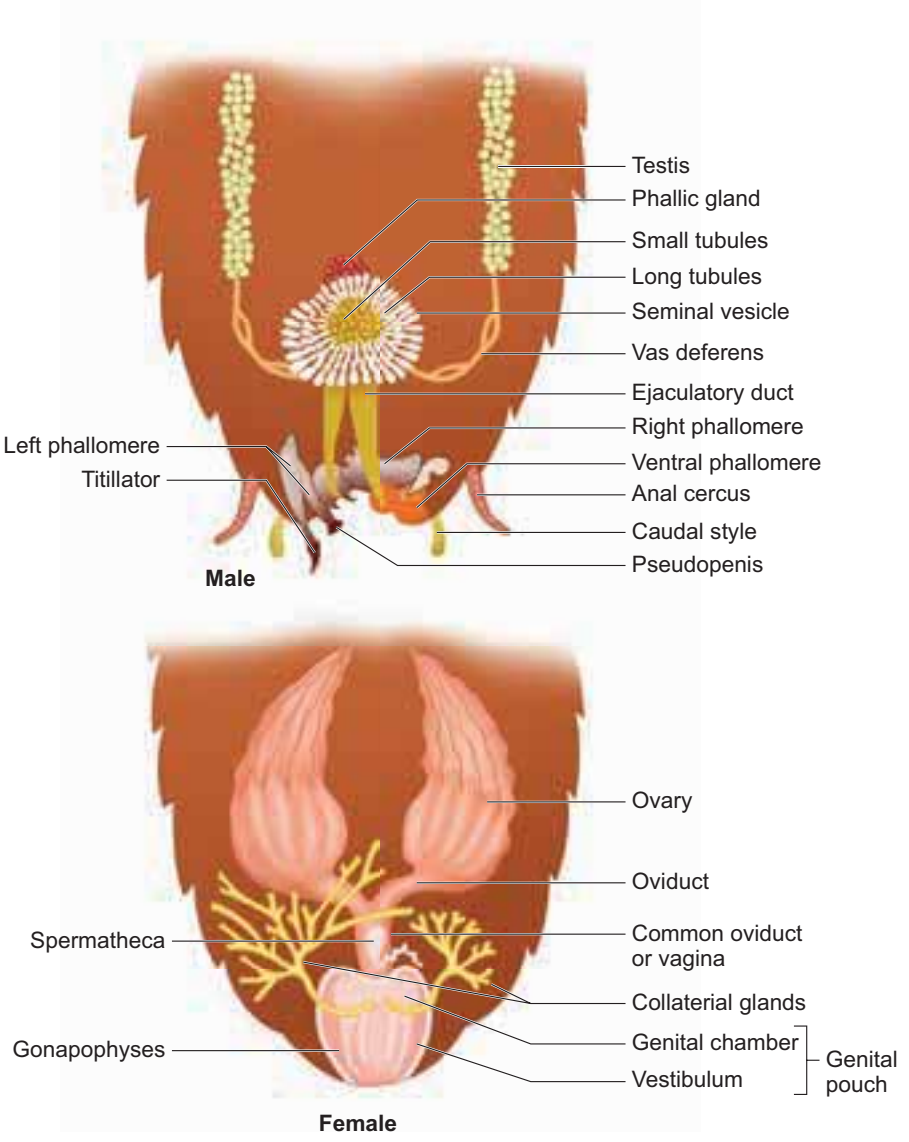


Figure 4.14 *Periplaneta americana reproductive system*